A. General Information

| Information Requested | Information provided by the PI |
|---|---|
| 1. **Heritage Craft Name** | *Muslin Fabric* |
| 2. **Affiliated Institute** | *ICAR-NINFET* |
| 3. **Principal Investigators** | *Dr. Santanu Basak*<br>*Dr. Manik Bhowmick*<br>*Dr. Sanjoy Debnath* |
| 4. **Official Project Title** | ***Technical Inputs in Handholding Muslin Weavers for Reviving Its Lost Glory*** |

## 1. Introduction

- **Brief description of the textile product**

Muslin, once celebrated as the "woven wind" or "fabric of the wind" for its extraordinary fineness, lightweight, transparency and breathability, represents one of the greatest textile heritages of the Indian subcontinent, particularly Bengal. Muslin cloth is made of cotton fabric, having plain weave. A minimum of 300-count yarn was used for the muslin, making it soft and transparent. It is very fine, light weight, comfortable fabric that can even pass through a metal ring. During the Mughal period and later in international trade with Europe, Bengal Muslin was regarded as a luxury fabric, admired for its softness, lightweight texture, and unmatched craftsmanship. However, colonial industrial policies, loss of traditional skills, and absence of technological modernization led to the gradual decline of this glorious heritage. Bengal Muslin was recognized as a Geographical Indication (GI) product of the Indian state of West Bengal in 2024.

Now, Muslin was produced in different regions of the Bengal subcontinent. Shantipur and Murshidabad district of Bengal mainly develops Muslin saree. Muslin was also produced in Malda and Hooghly district. The muslin produced at Sonargaon and its surrounding areas was of excellent quality, which is popularly known as Dhaka Muslin. The muslin produced in

Shantipur came to be known as Shantipuri Muslin, which was recognized by the East India Company. Muslin was made in Dhaka (Sonargaon) from very fine yarn, which is made from cotton called Phuti karpas; while in Malda, Radhanagar and Burdwan, muslin was made from fine yarn made from Suvin cotton, available in Nagpur and Guntur Districts of India. It is a heritage of Bengal. Now both the central and the state government take initiatives to revive its glory and persisting its heritage.

- **Geographical area of production (map if possible)**
  - It was produced mainly in Sonargaon(Dhaka) and its surroundings(currently in Bangladesh)
  - Now, In India muslin fabric is produced Santipur, Murshidabad, Malda, Hoogly and Bardhaman district of West Bengal

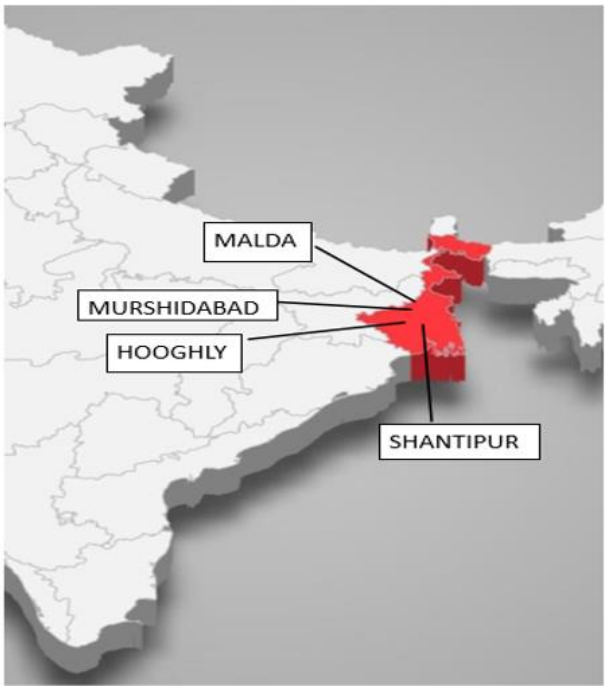


Fig 1. Different Muslin Fabric production zones of West Bengal, India

- **Key characteristics (material, technique, motifs, end-use)**
  - 100 % Cotton fibre having staple length more than 40 mm used for processing
  - As fine fabric that a whole fabric can pass through a ring

- Very few fibres (5-10) in yarn cross-section as Charka based hand spinning process
- Hand Ginning and Weaving in Handloom
- Generally no such signature motifs or patters are used.
- Muslin fabrics are mostly plain(1/1) in weave and white in color
- Shirt, Saree, Dhuti, Kurta- Kurti are generally produced from muslin fabric as an end product.

- **Why is it classified as a *textile heritage***

  Muslin cloth is classified as a textile heritage as of its historical significance from 17th to 18th centuries as Mughal Bengal emerged as the foremost muslin exporter in the world. Muslin fabric which is soft, lightweight, fine delicate fabric produced by the excellent craftsmanship of Bengal artisans. In the latter half of the 18th century, muslin weaving ceased in Bengal due to cheap fabrics from England and oppression by the colonialists. In India in the latter half of the 20th century and in Bangladesh in the second decade of the 21st century, initiatives were taken to revive muslin weaving. Bengal Muslin received Geographical Indication (GI) recognition for Indian state of West Bengal in 2024.Now, it is considered as local Heritage of India.

- **Cultural and traditional significance**
  - In royal courts of Mughal Empire Muslin symbolized Elegance, Refinement & Elite lifestyle.
  - It is a symbol of regional cultural identity and pride of India
  - Muslin represents Traditional Weaving Skills, Local Craftsmanship along with its Purity and simplicity.

- **Current market presence (local/national/export)**
  - World famous Bengal Muslin cloth is now experiencing a demand surge in export market mainly in the North American and European counties due to its sustainability.
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## 2. History

- **Origin of the craft (region/community)**
  - Muslin cloth was originated around 1000 BCE in the Bengal subcontinent of India.
  - It was produced mainly in Sonargaon (Dhaka) and its surroundings (currently in Bangladesh) and Santipur, Murshidabad (in west Bengal).
  - It reached its peak glory during 17-18th Century in Mughal Period.
  - Due to import of cheap fabrics from England and oppression by the colonialist, it lost its significance in British Colonial rule.

- Historical development

- **Major transformations over time (materials, techniques, designs)**

  In ancient period girls of age group 15 to 20 used to prepare Muslin yarn by hand. No machinery intervention was there. Total beauty of craft was dependent on skill and softness of hand. However, as of communication with Clusture, in 1980 also, they are familiar with Ambar Charka for Muslin yarn spinning, however with time machine setting, parameters have changed for better processing and production. In modern days, solar power, DC motor system driven are indulged in spinning system of Muslin yarn. However, in handloom weaving incorporation is Motor is strictly prohibited due to Heritage craft and GI intervention. Detail design of charka and other machines used in Muslin sector with time line will provide in due course

- **Timeline (mandatory): Ancient → Colonial → post-independence → Present**

  Muslin, once celebrated as the "woven wind" or "fabric of the wind" for its extraordinary fineness, lightweight, transparency and breathability, represents one of the greatest textile heritages of the Indian subcontinent, particularly Bengal. Muslin cloth is made of cotton fabric, having plain weave. A minimum of 300-count yarn was used for the muslin, making it soft and transparent. It is very fine, light weight, comfortable fabric that can even pass through a metal ring. During the Mughal period and later in international trade with Europe, Bengal Muslin was regarded as a luxury fabric, admired for its softness, lightweight texture, and unmatched craftsmanship. However, colonial industrial policies, loss of traditional skills, and absence of technological modernization led to the gradual decline of this glorious heritage. The muslin

produced at Sonargaon and its surrounding areas was of excellent quality, which is popularly known as Dhaka Muslin and it was very much popular throughout world in pre independence era. In those days, girls having age group of 15 to 20 are normally associated with this heritage craft. Muslin was made in Dhaka (Sonargaon) from very fine yarn, which is made from cotton called Phuti karpas. Recently Government of India and Government of West Bengal have taken various steps, schemes and initiaves to upgrade this heritage craft. Very recently, Bengal Muslin was recognized as a Geographical Indication (GI) product of the Indian state of West Bengal in 2024.

- Influence of government policies, industrialization, or globalization

3. Manpower

- **Number of families/artisans currently involved**
  500 (REGISTERED By MSME, khadi

- **Comparison with past (10–20 years ago)**

  20 years ago, almost 100-150 registered Muslin weavers were present in Bengal. Every year increasing by around 5 to 10% and currently 500 registered Muslin weavers are present in Bengal

- **Gender-wise and age-wise participation**

  Normally, women workers of age group 30 to 60 are associated with this profession

- **Skill levels (master artisans, helpers, trainees)**

  10 to 12 (lack of Master trainer in this area)

- **Average income per household (range)**
  Rs. 6,000 to Rs. 7000 per month normally

  Moreover, West Bengal Government provide Market Development Announce: 10 % of total production/ person in month basis + 10 % exgacia given by owner at the end of year on total production + MMDA given by central Govt 10 % of total production/ person in month basis

  With all these, total income of a person will be **Rs. 9000 to 10000 per month.**

- **Livelihood dependency (% income from this craft)**

  50% dependent from this craft

- **Reasons for increase/decrease in workforce**

  - Efficiency of weaving muslin fabric is very less as it is difficult to weave as it's a delicate fabric. Time limitation of weave a Muslin fabric is 9 pm to 9 am. If it is beyond 9 am then a lot of yarn breakage problem faced by weavers
  - Daily wage of spinners and weavers are very less of around Rs. 200 after working 8 hours
  - Ergonomical problem for spinners is one of the major issue
  - Due to high price, demand among long range of customers is not up to the mark
  - Normally Women's of age group 30 to 50 are associated with this profession as soft and expert hand is required for spinning of Muslin yarn
  - Lack of Master trainer and insufficient awareness and training programme to weavers and spinners

- **Migration trends (if any)**

  - Due to lack of profit in this area, skill labors are move in the other profession
  - Ergonomically difficult to spin and weave Muslin yarn and fabric therefore, associated workers are losing their interest from this profession
  - They are losing interest as daily wages are very low and it gives immense pressure on the artisan to meet their basic life needs.
  - Also losing interest as production rate is very low as the efficiency of muslin weaving is very low due to maintaining fabric delicacy as preparing one saree used to take 5 to 6 months.

- **Key challenges faced by artisans**

  - Low daily wages give pressure on the artisan to meet their basic life needs.
  - Marketing challenges
  - Ergonomically difficult to spin and weave Muslin yarn
  - Time limitation of weave a Muslin fabric is 9 pm to 9 am. If it is beyond 9 am then a lot of yarn breakage problem faced by weavers

- Lack of Master trainer and insufficient awareness and training programme to weavers and spinners

- **Expectations/aspirations of the community**

  - Different support from government
  - Technical support from government for technological up gradation as
  - Different Training program for skill development

**4. GI tag**

- **Why GI tag given**

  - Very fine yarn (count around 300-500 Nm) are produced here along with a good tensile strength.
  - "Woven Air Texture" is another reason for the GI tag. This means the fabric's transparency and very light weight (30-40 Gram per square meter)

- **Unique features**

  - Very fine fabric.
  - A whole fabric can pass through a ring
  - Simple 1/1 plain weave
  - Made of pure cotton fibre
  - Mostly white in colour
  - very soft in hand feel

- **Specify the GI Tag document no**. Banglar Muslin Certificate No. 564, Application No. 676

  Certificate Date: 1.03.2024, Valid up to 17.11.2029

## 5. Manufacturing steps

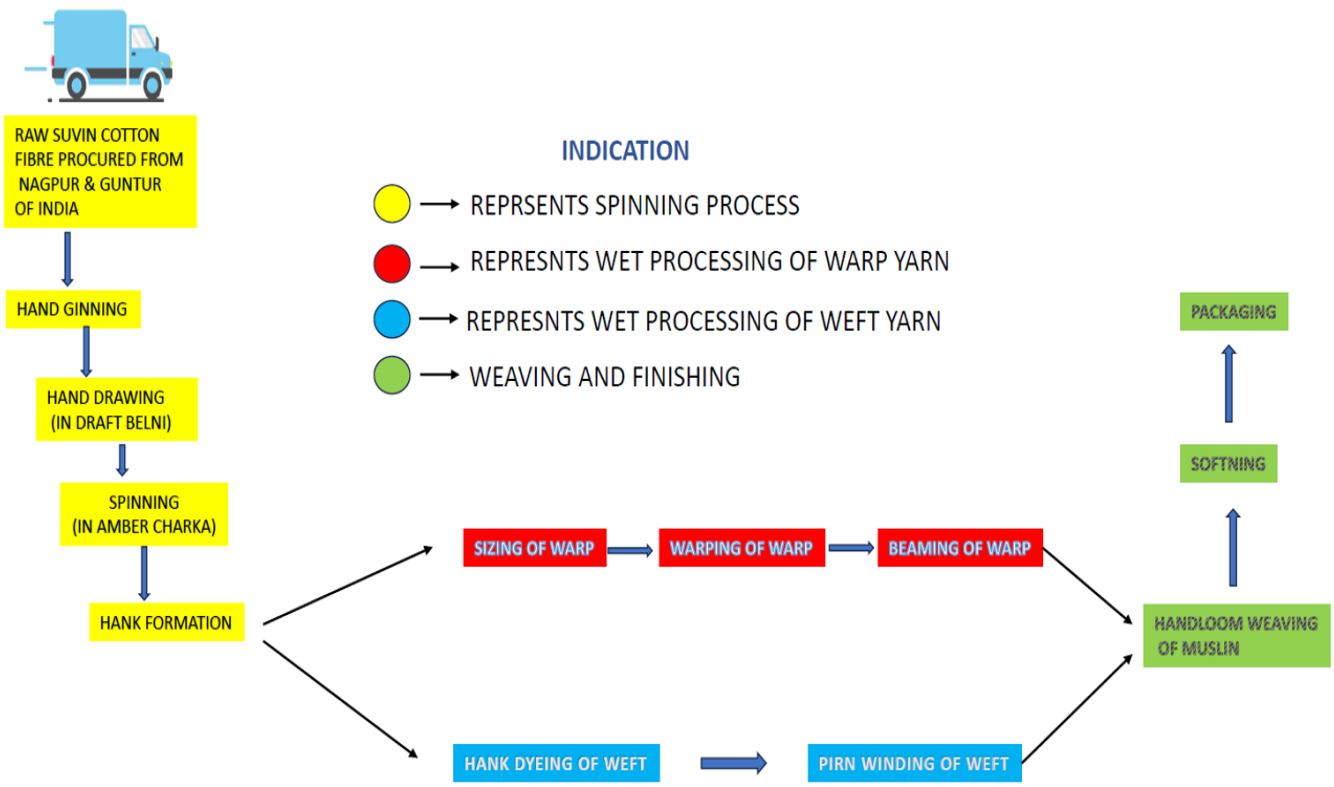


Fig 2. Detail; Manufacturing process of Muslin fabric with line diagram

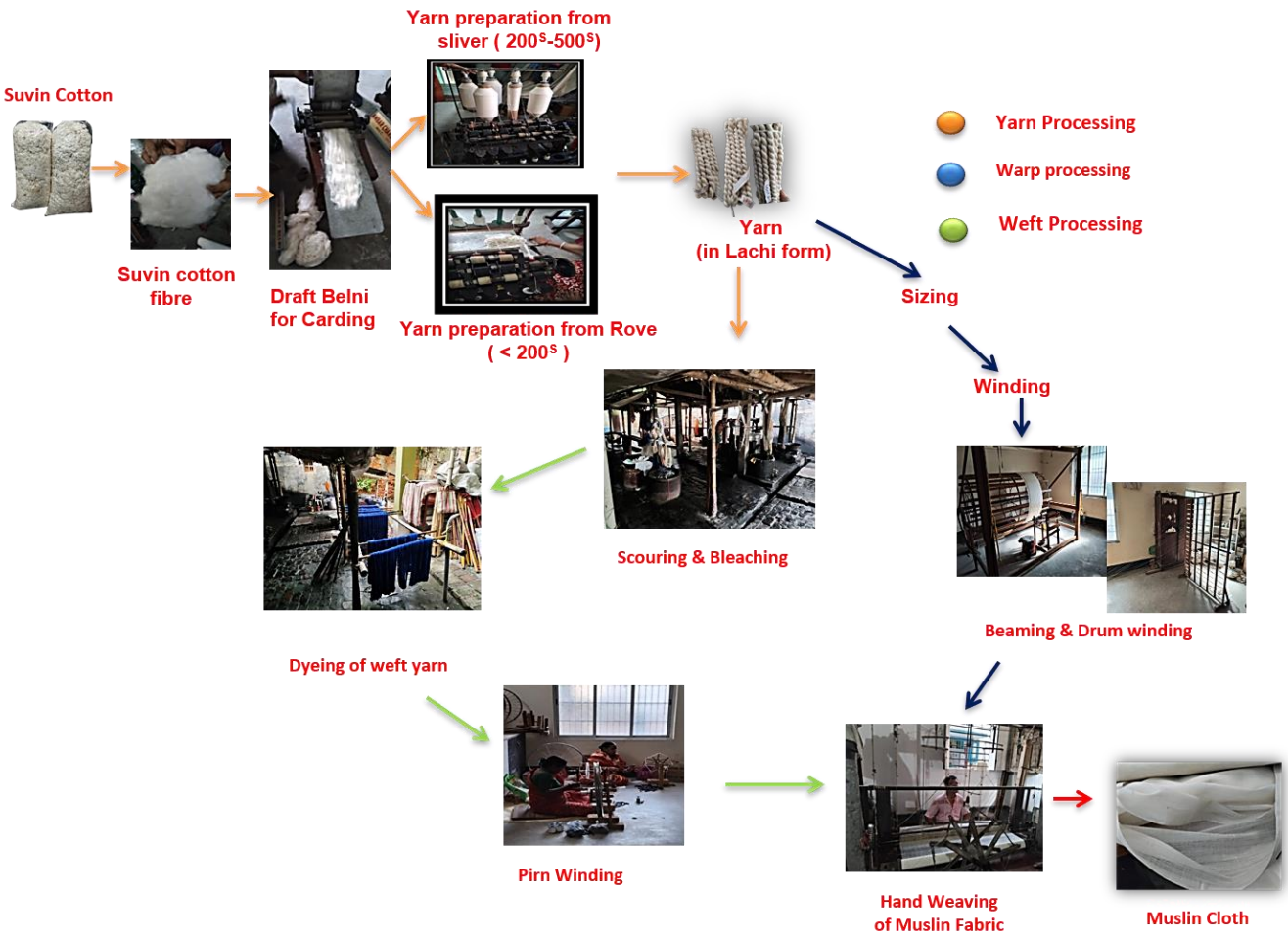


Fig 3. Detail; Manufacturing process of Muslin fabric with pictorial depiction

- **How much raw Materials are used per saree**

250 gm to 300 gm raw Muslin yarn required for making one saree if its prepared by **$180^S$ Ne** warp and weft yarn having count **$120^S$ Ne**

**Answer**

Length of saree 6.5 meter

Width of saree: 1.2 meter

Tested EPI: 80 & PPI: 120

Tested yarn count for wefp: 200 Nm ($120^S$ Ne) & warp: 300 Nm ($180^S$ Ne)

Total weft yarn required for this saree: (32 x 120x 650x5)/ 1 Lakh= 125 gm

Total warp yarn required for this saree: (47 x 120x 650x 3.2)/ 1 Lakh= 117 gm

Total Weight of yarn required for saree= (125+117) = 242 gm and the final calculated GSM of the material= 242/ (6.5x 1.2) = 31

Almost 130 Latchi/ hanks (**350$^{S}$** Ne) required for making one Muslin saree. One hank is 1000-meter length in size

- **Where the raw materials are sourced**- Nagpur and Guntur districts of India
- **How much energy is utilised in each process**

No energy is required for ginning, carding, spinning and weaving process as all of these process is traditional and carried out by hand

Only wet process required energy, however, they have used bleaching powder-based pretreatment for improving whiteness of material, therefore don't require any kind of heat energy for pretreatment.

Heat energy required for dyeing of Muslin yarn with Reactive dye (Cold brand)

Total weight of yarn for dyeing: 1.5 Kg

MLR used for Dyeing is: 1: 30

Amount of water required for dyeing: 45 Lt

Heat energy required for pretreatment: 45000 x 70 = 3150 Kcal

- **Form of energy- wood/coal/solar**

Some muslin weaving unit of west Bengal use solar energy for spinning process in Ambar charka to produce yarn from suvin cotton (available from Nagpur) fibre and LPG Gas for dyeing of weft yarn

6. Real photos of Cluster /Product/manufacturing process (Geo tagged if possible)

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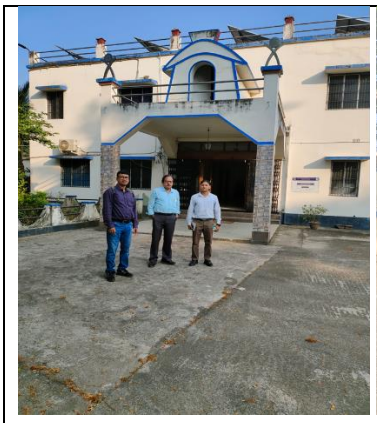
|  | |
| Visit at CoE, Khadi Unit, Ashok Nagar, Uttar 24 Pargana, West Bengal<br>17.11.2025 | Visited Matiary Kutir Shilpa for Muslin Yarn production at Nadia, West Bengal<br>12.09.2025 and 7.08.2025 |
| | |
| Visited Muslin Marketing Unit, Dankuni, Howrah on 21.04. 2025 | Visited weaving Unit of Muslin Fabric, Nadia West Bengal at 7.08.2025 |
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| Muslin Process unit Visit at Nadia, West Bengal on 11.10.2025 | Warping of Muslin Yarn at Ashok Nagar, Utttar 24 Pargana, West Bengal 17.11.2025 |

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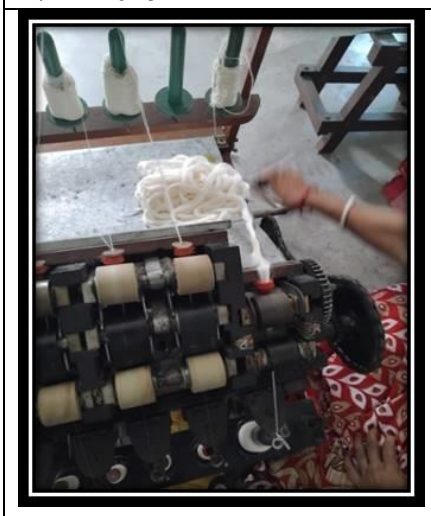
| Drum Winding of Muslin warp yarn on 17.11.2025 | Muslin yarn formation in Ambar Charka from Rove sliver |
|  | |
| Muslin yarn formation in Ambar Charka from carded sliver | Feed material of cotton after hand ginning/ hand carding of cotton fibre in Draft Belni machine |
| | |
| **Training of Muslin yarn Pretreatment and Dyeing conducted at Ashoknagar from 27.01.2026 to 31.01.2026** | |
| | |
| Training Delivered at Ashoknagar from 27 to 31.01.2026 | Training of peroxide bleaching process |

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| | |
| Dyeing of Muslin yarn with Ramazol colour by following hand dyeing process | Pretreatment process performed by trainee members at Ashoknagar |
| | |
| Inaugural session and discussion with trainee members at Ashoknagar on 27.01.2026 | Certificate distribution to trainee members at Ashoknagar on 31. 01.2026 |

**Training of Muslin yarn Pretreatment and Dyeing conducted at Thiba Gramin and Anchalik Welfare Society, Lavpur, Birbhum, west Bengal from 16.02.2026 to 20.02.2026**

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| Training delivered at Thiba, Birbhum, West Bengal<br>On 16.02.2026 | Valedictory session of the training programme on 20.02.2026 |
| | |
| Certificate distribution to trainee members at Thiba, on 20.02.2026 | Direct dyeing of Muslin yarns performed by trainee members |

7. Videos (Short clips of weavers describing their journey)

## 8. Technological Intervention

- No standard scientific documentation of spinning parameters and fabric properties. So proper documentation of all spinning parameters, fabric and yarn properties are required

- Increasing weaving time period (by adding humidification system) and validation of the process in different Muslin units

- Lack of standard bleaching and dyeing techniques used so clusture need to be improved with modified bleaching process and other dyeing (Ramazol Reactive dyeing, Direct dye for coarser Muslin yarn products) techniques

- In wet processing part, production need to be improved through DC power/ Solar power based machinery intervention. (Challenge of Fabric dyeing).

- Sizing is manual and mundane process and intervention with some machinery set up will be Advantageous in this area.

- As of verbal communication with workers associated with the Muslin field, ergonomical improvement of spinners are very much required and one of the important vacant space to work with.

- Different awareness programmes and workshops are required through MSME for Heritage revival.